

Computer Networks: Subnetting Fundamentals & FLSM Workflow

Comprehensive Study Notes

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1 What is Subnetting?

Subnetting is the process of taking a single physical network and logically dividing it into smaller, manageable sub-networks (**Subnets**).

How Subnetting Works: Bit Borrowing

An IPv4 address consists of 32 bits divided between the **Network Portion** and the **Host Portion**.

When subnetting a network, the original **Host Portion** is split into two distinct parts:

1. **Borrowed Subnet Portion:** Bits taken from the original host bits to uniquely identify individual subnets.
2. **Remaining Host Portion:** Bits left over to identify specific devices (hosts) within each subnet.

$$\underbrace{\text{Original Network Bits}}_{\text{Inherited Net ID}} + \underbrace{\text{Borrowed Subnet Bits}}_{\text{Subnet Portion}} + \underbrace{\text{Remaining Host Bits}}_{\text{Host Portion}} = 32 \text{ Bits}$$

2 Infrastructure IP Address Requirements

Understanding which network elements require IP addresses is fundamental to accurate network design:

Core Hardware IP Rules

- **Switches (Layer 2):** Do NOT require an IP address for standard frame switching and forwarding functions within a local area network (LAN).
- **End Devices:** Every end device (PCs, laptops, servers, printers, smartphones) connected to the network requires a unique IP address.
- **Router Interfaces (Layer 3):** Each interface on a router is considered a host on its connected network. Every router interface connects to a distinct subnet and must be assigned a unique IP address within that subnet's range.

3 Subnetting Methodologies

There are two primary methods for subnetting an IP address block:

1. **FLSM (Fixed Length Subnet Mask):** The traditional method where **all subnets are allocated the exact same size** (same number of host bits and identical subnet mask), regardless of varying host requirements across different segments.
2. **VLSM (Variable Length Subnet Mask):** The modern method where subnets are customized with **different prefix lengths** based on exact host demands, optimizing IP address utilization and minimizing wasted addresses.

4 FLSM Mathematical Algorithm & Workflow

To divide a network block using **FLSM** based on the largest host count requirement (H hosts), follow this exact mathematical procedure:

Step 1: Find the Largest Subnet Requirement (H): Identify the segment that requires the highest number of hosts (e.g., $H = 60$ hosts).

Step 2: Account for Network & Broadcast Addresses (H_{total}): Add 2 addresses to the required host count (1 for Network IP + 1 for Broadcast IP):

$$H_{\text{total}} = H + 2 \implies 60 + 2 = \mathbf{62 \text{ IPs}}$$

Step 3: Calculate Remaining Host Bits (h): Take the ceiling of the base-2 logarithm of H_{total} :

$$h = \lceil \log_2(H_{\text{total}}) \rceil = \lceil \log_2(62) \rceil = \mathbf{6 \text{ bits}}$$

(Explanation: $2^5 = 32 < 62 \leq 2^6 = 64$, so 6 host bits are required.)

Step 4: Calculate New Network Prefix Length (P_{new}): Subtract the remaining host bits from the total 32 bits:

$$\text{Net Bit Count } (P_{\text{new}}) = 32 - h = 32 - 6 = \mathbf{/26}$$

Step 5: Calculate Total Allocated IPs per Subnet (Block Size):

$$\text{Allocated IPs} = 2^h = 2^6 = \mathbf{64 \text{ addresses}}$$

Step 6: Calculate Borrowed Subnet Bits (s): Subtract the new host bits from original host bits, or subtract original prefix from new prefix:

$$\text{Borrowed Bits } (s) = \text{Original Host Bits} - \text{Subnet Host Bits} = P_{\text{new}} - P_{\text{original}}$$

Step 7: Calculate Subnet Mask: To calculate the new subnet mask, evaluate the binary mask using **only the remaining host bits after FLSM** (i.e., $h = 6$ host bits set to '0', and 26 network bits set to '1'):

$$\text{Last Octet Value} = 256 - \text{Block Size} = 256 - 2^h = 256 - 64 = \mathbf{192}$$

$$\text{Subnet Mask} = \mathbf{255.255.255.192}$$

5 Worked Example Walkthrough

Problem: Given the base network **192.168.10.0/24**, create subnets using FLSM to support a maximum requirement of **60 hosts**.

Step-by-Step Resolution

- **Base Network:** 192.168.10.0/24 (Original Host Bits = $32 - 24 = 8$)
- **Largest Host Count (H):** 60
- **Total IPs Required (H_{total}):** $60 + 2 = 62$
- **Required Host Bits (h):** $\lceil \log_2(62) \rceil = 6$ bits
- **Borrowed Subnet Bits (s):** Original Host Bits $- h = 8 - 6 = 2$ bits
- **New CIDR Prefix (P_{new}):** $32 - 6 = /26$
- **Allocated IPs per Subnet (Block Size):** $2^6 = 64$
- **Total Subnets Created:** $2^s = 2^2 = 4$ Subnets
- **Subnet Mask:** $255.255.255.(256 - 64) = 255.255.255.192$

5.1 FLSM Subnet Allocation Table

Table 1: FLSM Subnet Breakdown for 192.168.10.0/24 (60 Hosts Requirement)

Subnet ID	Prefix	Network IP	Usable Host Range	Broadcast IP	Subnet Mask
Subnet 0	/26	192.168.10.0	192.168.10.1 – 10.62	192.168.10.63	255.255.255.192
Subnet 1	/26	192.168.10.64	192.168.10.65 – 10.126	192.168.10.127	255.255.255.192
Subnet 2	/26	192.168.10.128	192.168.10.129 – 10.190	192.168.10.191	255.255.255.192
Subnet 3	/26	192.168.10.192	192.168.10.193 – 10.254	192.168.10.255	255.255.255.192